Code-Layout, Readability and Reusability

Date	02 October 2026
Team ID	SWTID-2026-3981
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	PEP 8 coding standard follow out standard 4-space indentation use.
2	Proper File Structure	Files and folders are logically organized	Yes	Routes, templates, models, and static assets separate directories lo arrange.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Clear naming conventions (e.g., user_prompt, quiz_questions, gemini_api_key) use.
4	Function / Method Names	Functions are descriptively named	Yes	Functions descriptive (e.g., generate_quiz(), summarize_text(), get_learning_path()) maintain.
5	Code Comments	Inline and block comments explain logic	Yes	Doc strings and inline comments API setup and local model loading clearly explain.
6	Modular Design	Code is split into reusable functions/modules	Yes	Fast API routes and AI model inference functions modular

				functions separate.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate or unused code is removed.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Exceptions and errors are handled gracefully.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Gemini Model Wrapper Service	Python / google-generativeai	Q&A,Summarization, Quiz Generation,and Recommendation routes reuse.	High
2	Offline Local Inference Pipeline	Python /PyTorch, Transformers	Local fallback Q&A processing, local concept explanation modules reuse.	High
3	Response Formatter & JSON Parser	Python /Pydantic,JSON	API responses,structured quiz formatting, and UI template rendering reuse.	High
4	UI Template Base Layout	HTML / Jinja2 Templating	Web app dashboard pages, quiz displays,and summary views.	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5/5	FastAPI architecture standard Python project directory layout strictly follow.
Readability	5/5	PEP 8 conventions,meaningful variables/function naming follow readability.
Reusability	4/5	Model inference logic and UI components modular structures organize.
Documentation / Comments	4/5	Functions and API routes in comprehensive docstrings and clear logic comments.
Overall Score	4.5/5	Highly maintainable,salable,and modular code base.